



# Breathing Life Into the Forgotten:

How Artificial Intelligence is Transforming  
Tuberculosis Screening in Rural India

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**2022 → 2023**

Pilot Data

# The Crisis That Hides in Plain Sight

Tuberculosis remains the world's deadliest infectious disease — and India bears 27% of the global burden.



## 2.8M

New TB Cases per Year in India



## 3 Days

Average Diagnosis Time (Before AI)



## 78%

Detection Accuracy Before AI

*Each undiagnosed carrier infects an average of 10–15 people before treatment begins.*



*Earlier, when I found someone who was coughing badly, I had to tell them to come to the health centre and wait. Many of them did not come back. Now I can show them something on the screen the same day.*

**Sunita Bai**

ASHA Worker, Jharkhand

#### KAMLA DEVI'S STORY

*A 34-year-old agricultural labourer in Madhya Pradesh*



#### 5 Months Undetected

Cough dismissed as dust or cold. No accessible clinic.



#### 11 km to Nearest Clinic

Unpaved track, broken equipment, no X-ray.



#### 3-Day Wait for Results

Lab results from city — already infected 2 children



#### 2023: Screened by AI Unit

Mobile team, same-day result, treatment started.

*"By the time she was diagnosed, she had been ill for nearly five months."*



# The AI-Powered Screening System

*A portable, intelligent, last-mile diagnostic pipeline — deployed on a jeep, diagnosing lives.*

**01**



## Community Outreach

ASHA/ANM workers identify symptomatic individuals in remote villages.

**02**



## Portable X-Ray

Digital chest X-ray taken on mobile equipment loaded on a jeep.

**03**



## AI Analysis

Deep-learning model trained on 1M+ images scans X-ray in <12 seconds.

**04**



## Doctor Review

Flagged cases reviewed remotely by a physician via telemedicine.

**05**



## Same-Day Result

Patient receives diagnosis and initiates treatment within 12 hours.

*The AI acts as a second pair of expert eyes — not replacing doctors, but multiplying their reach across 700,000 villages.*

## DATA SNAPSHOT

# 2022 (Before AI) vs. 2023 (After AI)

### PATIENTS SCREENED

BEFORE

45,000

AFTER AI

82,000

+82%

### DETECTION ACCURACY

BEFORE

78%

AFTER AI

92%

+14%

### AVG. DIAGNOSIS TIME

BEFORE

3 Days

AFTER AI

12 Hrs

-83%

### FALSE NEGATIVES

BEFORE

230

AFTER AI

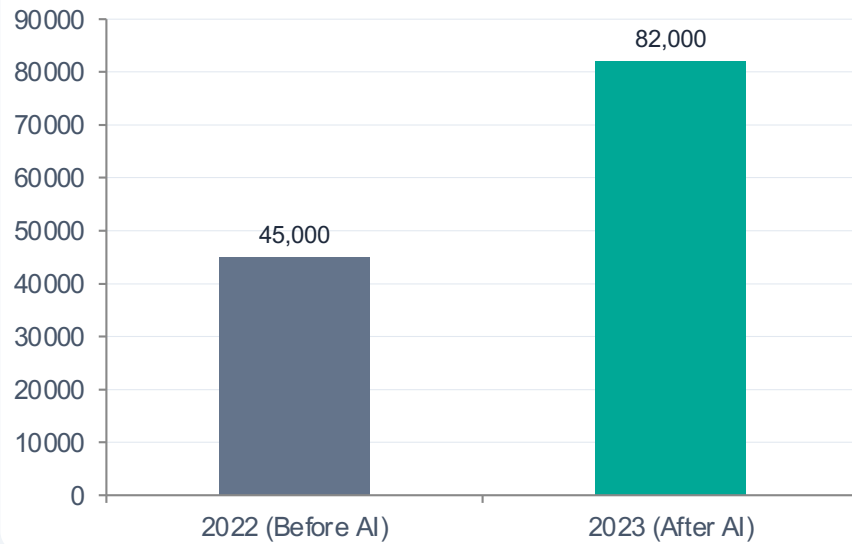
85

-63%

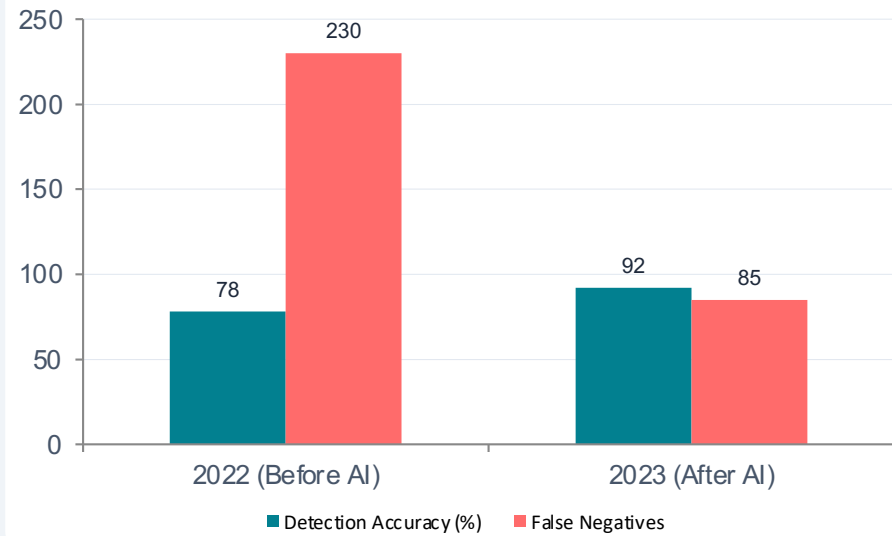
*Testimonials collected from field health workers and patients across pilot districts.*

# Impact at a Glance

Patients Screened



Accuracy & False Negatives



**+82%**  
More patients reached

**-83%**  
Faster diagnosis

**-63%**  
Fewer missed cases



# POLICY IMPLICATIONS

What this means for India's National TB Elimination Programme and policymakers at every level.

*"Prevention through early detection is not charity — it is the most cost-effective public health investment available."*



## Scale Without Specialists

AI reduces dependency on concentrated radiologist resources. Portable diagnostics deliver specialist-grade accuracy to Tier-3 and Tier-4 health centres.



## Equity as Efficiency

The system improves outcomes precisely where infrastructure is weakest. The urban-rural diagnostic gap narrows when diagnosis becomes portable.



## The Economics Are Clear

Early-stage TB treatment costs Rs 5-8K. Drug-resistant TB exceeds Rs 1.5 lakh. Every missed diagnosis compounds cost exponentially.



## A Scalable Architecture Exists

1M+ ASHA workers, proven AI models, and portable X-ray equipment form a ready deployment infrastructure for national rollout.

THE ROAD AHEAD

# From Pilot to National Architecture

## PHASE 1

2024-2025

### Expand the Pilot

- ✓ Scale from pilot districts to 100 high-burden districts
- ✓ Train 50,000 ASHA workers on tablet interface
- ✓ Procure 500 portable digital X-ray units

## PHASE 2

2025-2027

### Build the Backbone

- ✓ Establish state-level AI diagnostic hubs
- ✓ Integrate with Ayushman Bharat digital health records
- ✓ Deploy telemedicine networks in all Tier-3 centres

## PHASE 3

2027-2030

### National Elimination Push

- ✓ Universal AI-assisted screening for all symptomatic patients
- ✓ Real-time epidemiological dashboard for MoHFW
- ✓ India as global model — export framework to 22 TB-endemic nations

India's National TB Elimination Target: 2030 | AI-assisted screening is the acceleration mechanism, not a replacement for political will.



# The Economic Case for Investment

*Early detection is not charity — it is the most cost-effective public health decision available.*

## EARLY DETECTION (AI-Assisted)

# ₹5,000–₹8,000

Per patient, 6-month drug-sensitive TB treatment

- ✓ 95% cure rate when caught early
- ✓ Minimal hospitalisation required
- ✓ Community transmission halted quickly
- ✓ Patient maintains economic productivity

## LATE DETECTION (Drug-Resistant TB)

# >₹1.5 Lakh

Per patient, complex MDR-TB treatment with hospitalisation

- ✗ Lower cure rates, longer treatment duration
- ✗ Hospital bed occupation for months
- ✗ 10–15 additional community infections
- ✗ Household economic collapse, children withdrawn from school

**Every ₹1 invested in AI-assisted early TB screening generates an estimated ₹15–20 in avoided treatment costs and productivity gains.**

## CONCLUSION

# The Machine Found Her in Time.

- 82,000 patients screened — an 82% increase — through mobile AI units in a single year.
- Detection accuracy rose to 92%, reducing missed cases that drive community transmission.
- Diagnosis time fell from 3 days to 12 hours — keeping the 95% cure-rate window open.
- The technology is proven. The infrastructure exists. The policy decision awaits.

The question for policymakers: not whether this works — but how fast we make it universal.



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